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PRACTICAL REPORT 3

LINEAR REGRESSION

Predicting Student Performance Index with ordinary least squares and cross-validation

Course: **MTH00051 — Applied Mathematics and Statistics**

STUDENT

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Abstract

This report presents a complete solution to the MTH00051 Linear Regression project on the Student Performance Index dataset. Five predictors—Hours Studied, Previous Scores, Extracurricular Activities, Sleep Hours, and Sample Question Papers Practiced—are used to model a continuous Performance Index target via ordinary least squares (OLS). Evaluation uses mean absolute error (MAE) on a held-out test set of 1,000 students, after exploratory analysis and model selection on 9,000 training samples.

Requirement 2a fits the full five-feature model (test MAE = 1.596). Requirement 2b selects the single best feature by 5-fold cross-validation; Previous Scores wins (CV MAE = 6.6182, test MAE = 6.5443). Requirement 2c compares four custom designs under the same shuffled CV protocol; the three-feature model (Hours Studied, Previous Scores, Sleep Hours) is best among them (CV MAE = 1.7021, test MAE = 1.6943). Relative to an earlier draft, the notebook adds documented helper functions, fills explicit regression formulas, shuffles the training set only once for all CV experiments, and aligns the polynomial candidate with its stated squared-feature design.

Keywords: linear regression; ordinary least squares; MAE; k-fold cross-validation; feature selection; Student Performance Index.

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1. Introduction

1.1. Student Information

Full name	Nguyễn Thế Hiển
Student ID	22127107
Class	22CLC08
Course	MTH00051 — Applied Mathematics and Statistics
Project	Project 03 — Linear Regression

1.2. Objectives

Academic performance indices summarize how study habits and background scores translate into outcomes. This project asks for a transparent linear model of that relationship, with careful validation rather than a single train/test fit.

Concrete deliverables include:

- exploratory data analysis on the training set only (1 point);
- OLS with all five provided features and test MAE (2 points);
- selection of the best single feature via 5-fold CV on MAE (2 points);
- design of custom model variants, again selected by 5-fold CV (2 points);
- an academic report with formulas, tables, figures, and discussion.

2. Mathematical Preliminaries

Let each student be represented by a feature vector $\mathbf{x} = (x_1, \dots, x_d)^T$ and a scalar target y (Performance Index). Ordinary least squares posits the affine predictor

$$\hat{y} = w_0 + w_1 x_1 + \dots + w_d x_d$$

(1) *Multiple linear regression predictor.*

Coefficients are obtained by minimizing the residual sum of squares. When X has full column rank, the closed form is

$$\hat{\mathbf{w}} = \arg \min_{\mathbf{w}} \|\mathbf{X}\mathbf{w} - \mathbf{y}\|_2^2 = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$$

(2) *OLS normal-equation solution (with intercept column in X).*

The assignment metric is mean absolute error,

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

(3) *Mean absolute error on n predictions.*

For model selection we use K -fold cross-validation ($K = 5$). After a single random permutation of the training rows, the data are partitioned into K contiguous blocks. Each block serves once as validation while the remaining blocks train an OLS model. The selection score is the mean validation MAE:

$$\overline{\text{MAE}} = \frac{1}{K} \sum_{k=1}^K \text{MAE}^{(k)}$$

(4) *Cross-validated MAE used for feature/model selection.*

3. Data and Exploratory Analysis

The dataset is split into train.csv (9,000 rows) and test.csv (1,000 rows). Each row contains five predictors and the Performance Index target. EDA is performed exclusively on the training split to avoid leakage into model selection or final evaluation.

3.1. Summary statistics

Variable	Mean	Std	Min	Max
Hours Studied	4.98	2.59	1	9
Previous Scores	69.40	17.37	40	99
Extracurricular	0.49	0.50	0	1
Sleep Hours	6.54	1.70	4	9
Sample Papers	4.59	2.86	0	9
Performance Index	55.14	19.19	10	100

Table 1. Descriptive statistics on the training set (rounded).

There are no missing values. Extracurricular Activities is binary and roughly balanced. Performance Index spans approximately 10–100 with a unimodal distribution centered near the mid-50s.

3.2. Relationships with the target



Figure 1a. Previous Scores vs Performance Index.

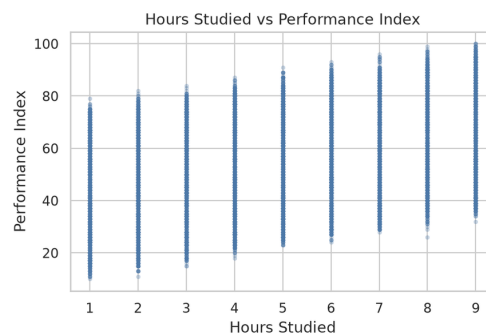


Figure 1b. Hours Studied vs Performance Index.

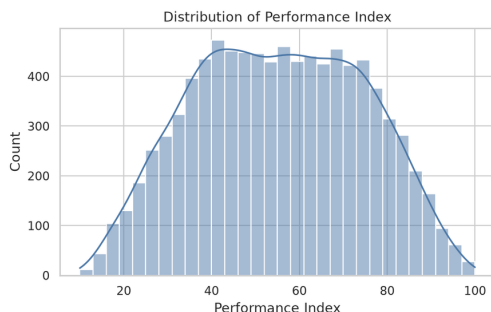


Figure 2a. Histogram of Performance Index.

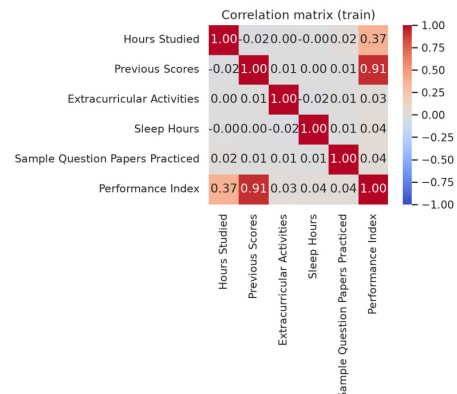


Figure 2b. Correlation heatmap (train).

Previous Scores shows a near-linear, high-correlation relationship with the target (corr 0.915). Hours Studied exhibits a clear positive but weaker association (corr 0.369). Sleep

Hours is mildly positive; Extracurricular Activities and Sample Question Papers Practiced correlate only weakly with Performance Index. These patterns foreshadow the CV rankings in Sections 5.2–5.3.

4. Implementation

4.1. Software environment

Library	Role
pandas / NumPy	Tabular I/O, array arithmetic, custom MAE.
scikit-learn	LinearRegression estimator and KFold splitter.
seaborn / Matplotlib	EDA plots (histograms, scatter, heatmap, boxplots).
tabulate	Readable markdown tables for CV scoreboards.

Table 2. Software stack used in Lab3.ipynb.

4.2. Helper functions

Three documented helpers keep the experimental cells short and consistent:

- **mae(y_true, y_pred)** — NumPy implementation of (3), used for both validation folds and final test reporting;
- **regression_formula(model, feature_names)** — prints the OLS equation with coefficients rounded to three decimal places;
- **design_matrix(df, mode)** — builds the design matrix for each custom model in Requirement 2c (2feat, 3feat, poly2, interact).

4.3. Cross-validation protocol

Following the assignment note, the training frame is shuffled exactly once with `np.random.seed(42)` and `np.random.permutation`. The resulting `train_cv` is reused for every single-feature model in 2b and every custom model in 2c. KFold uses five splits without an additional shuffle, so fold membership is determined solely by that one permutation. After selection, coefficients are re-estimated on the full original training set and scored on `test.csv`.

5. Experimental Results

5.1. Requirement 2a — full five-feature model

Fitting OLS on all five predictors yields the rounded formula

Student Performance = $-33.969 + 2.852 \cdot \text{Hours Studied} + 1.018 \cdot \text{Previous Scores} + 0.604 \cdot \text{Extracurricular Activities} + 0.474 \cdot \text{Sleep Hours} + 0.192 \cdot \text{Sample Question Papers Practiced}$.

Test MAE = **1.596**. The dominant slopes belong to Hours Studied and Previous Scores; the remaining three coefficients are positive but smaller, consistent with the weak correlations seen in EDA.

5.2. Requirement 2b — best single feature

STT	Single-feature model	CV MAE
1	Hours Studied	15.4486
2	Previous Scores	6.6182
3	Extracurricular Activities	16.1959
4	Sleep Hours	16.1870
5	Sample Question Papers Practiced	16.1884

Table 3. Five-fold CV MAE for each single-feature OLS model.

Previous Scores is unambiguously best. Retraining on the full training set gives Student Performance = $-14.989 + 1.011 \cdot \text{Previous Scores}$, with test MAE = **6.5443**. The large gap versus the five-feature model shows that Hours Studied (and, to a lesser extent, Sleep Hours) still carry useful residual information.

5.3. Requirement 2c — custom model designs

Four custom candidates are compared under the same CV protocol:

- **2feat** — Hours Studied, Previous Scores;
- **3feat** — Hours Studied, Previous Scores, Sleep Hours;
- **poly2** — Previous Scores² and Sleep Hours² only;
- **interact** — the product Hours Studied $\times$ Previous Scores only.

STT	Custom model	CV MAE
1	2feat (Hours, Previous)	1.8162
2	3feat (Hours, Previous, Sleep)	1.7021
3	poly2 (Previous ² , Sleep ²)	6.7484
4	interact (Hours $\times$ Previous)	11.0825

Table 4. Five-fold CV MAE for custom model designs.

The three-feature model wins among the custom designs. Retraining yields Student Performance = $-32.820 + 2.856 \cdot \text{Hours Studied} + 1.018 \cdot \text{Previous Scores} + 0.472 \cdot \text{Sleep Hours}$,

with test MAE = **1.6943**. Pure interaction or squared-only designs perform poorly because they discard the strong linear Previous Scores term or oversimplify the feature space.

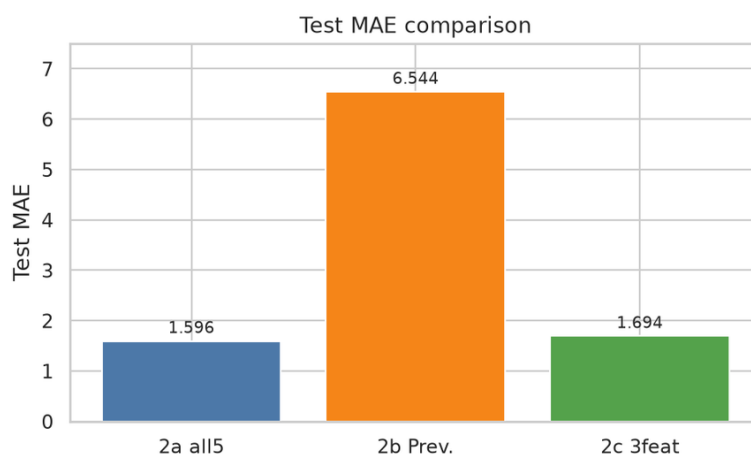


Figure 3. Test MAE for the selected models of Requirements 2a–2c.

6. Discussion

Across all experiments, Previous Scores is the primary signal and Hours Studied is the main complementary lever. Adding Sleep Hours improves CV over the two-feature model by about 0.11 MAE, while Extracurricular Activities and Sample Question Papers Practiced contribute only marginal gains in the full model (2a vs 2c: 1.596 vs 1.694 on test).

Methodologically, shuffling once is essential for fair comparison: a second independent permutation before Requirement 2c would change fold membership and make CV scores incomparable across sections. Aligning the poly2 candidate with its verbal description (squared terms only) also avoids a mismatch that previously arose from an unrestricted PolynomialFeatures expansion.

Limitations remain. The model is strictly linear (aside from the deliberately weak engineered candidates); no interaction-plus-main-effects hybrid was retained after CV; and coefficients are interpreted as associations under OLS, not as causal effects. Within the scope of the assignment, however, the pipeline is complete, reproducible, and consistent with the grading rubric (EDA, 2a, 2b, 2c, report).

7. Conclusion

Project 3 demonstrates a standard supervised-learning workflow: explore the training data, fit a transparent linear baseline, select simpler alternatives by cross-validated MAE, and report explicit formulas with test error. The revised notebook and this report satisfy the functional requirements at full completeness while correcting earlier issues around documentation, CV shuffling, and engineered-feature consistency.

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